Stereotactic Radiotherapy of Uveal Melanoma

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Contents

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15.1 Introduction

Stereotactic radiotherapy involves the precise positioning of the tumor in three-dimensional space by appropriate scanning and the delivery of ionizing radiation to the tumor from multiple directions [1]. A high dose of radiation is therefore focused at the target with relatively little irradiation of the surrounding healthy tissues.

This chapter is based upon contributions of independent investigators who jointly present an overview of this novel approach to the radiotherapy of uveal melanoma. The sections on radiosurgery and fractionated stereotactic radiotherapy, respectively, are written by workers from Graz and Vienna, both in Austria, with a contribution from British Columbia, Canada.

15.2 Types of Stereotactic Radiotherapy

There are two techniques of delivering stereotactic radiation to the eye: stereotactic radiosurgery and fractionated stereotactic radiotherapy.

15.2.1 Stereotactic Radiosurgery

Stereotactic radiosurgery (SRS) involves multiple radiation beams that are focused onto the tumor simultaneously from different directions, the treatment usually being completed in a single session or occasionally fractionated over several days.

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15.2.2 Fractionated Stereotactic Radiotherapy

During fractionated stereotactic radiotherapy (SRT), a single beam of radiation is aimed at the tumor successively from different directions, the entire treatment being delivered either in a single session or, as is usually the case, in a fractionated manner over several sessions (Box 15.1).

Box 15.1. Stereotactic Radiotherapy

- Ionizing radiation is delivered to the tumor from multiple directions either concurrently (stereotactic radiosurgery) or sequentially (stereotactic radiotherapy).
- The tumor is positioned precisely in three-dimensional space by appropriate scanning.
- The patient's head and eye are immobilized using a variety of methods.
- Stereotactic radiotherapy delivers a high dose of radiation to the tumor with small doses to the surrounding tissues.
- Early results are encouraging, indicating that this modality has a place in the treatment of uveal melanomas.
- These methods are used mostly for posterior tumors that are difficult to treat with plaque and for large tumors.

15.3 Stereotactic Radiosurgery

Stereotactic radiosurgery is minimally invasive and delivers a single focal dose of high-energy x-ray or gamma radiation to a stereotactically defined target while sparing the healthy surrounding tissues (Fig. 15.1).

15.3.1 Treatment Planning

Treatment plans are based on MRI scans. Gadolinium enhancement is used to distinguish the tumor (gross volume) from subretinal fluid. During treatment planning, critical structures such as the optic nerve, retina, macula, lens, and ciliary body are identified. A dose of 30 Gy is administered. A safety margin of 1–2 mm around the tumor is treated (planned treatment volume) (Fig. 15.2). Dose-volume histogram analysis is

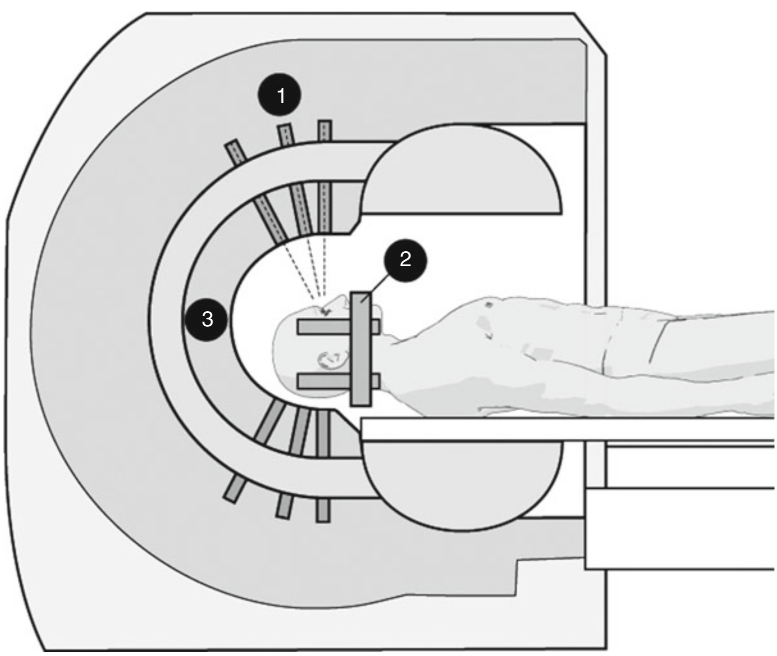

Fig. 15.1 Gamma Knife unit with main body containing Co-60 cources and primary collimator system (**1**), Leksell head frame (**2**), secondary collimator (**3**)

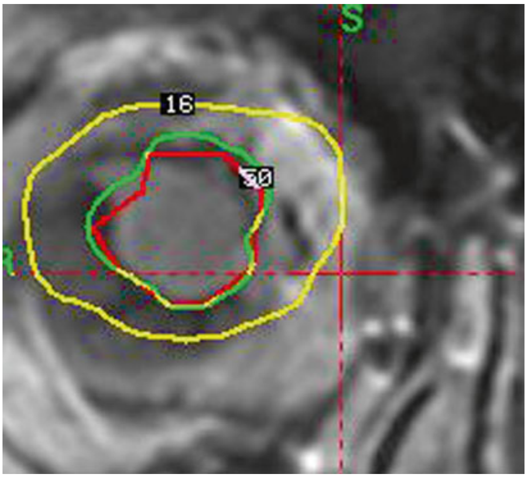

Fig. 15.2 Dosimetry plan for Gamma Knife treatment of an uveal melanoma, depicting CTV (red), PTV at 50 % isodose (*green*) and 16 % isodose (*yellow*)

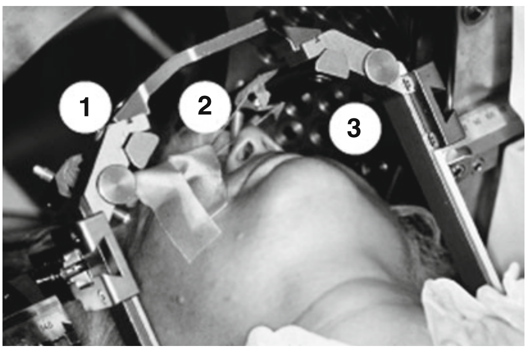

Fig. 15.3 A patient positioned within Gamma Knife prior to treatment. Leksell head frame (**1**), traction sutures (**2**), secondy collimator (**3**)

also performed to determine the likelihood of complications [2].

15.3.2 Procedure

The radiosurgical procedure is usually performed with the patient in the supine position. The eye is immobilized with a precision of ±0.3 mm, by combining a retrobulbar anesthetic block either with transconjunctival sutures placed in the rectus muscles [3] or with a vacuum device [4]. The patient is positioned with the back of the head very close to the frame. Pins are fixed to the supraorbital region to position the eye as close to the center of the stereotactic space as possible (Fig. 15.3).

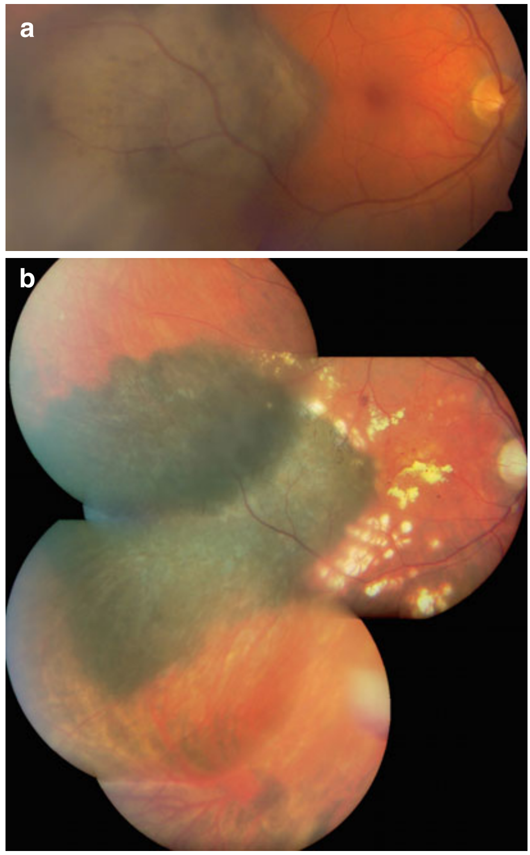

Fig. 15.4 Patient before (a) and after (b) Gamma Knife treatment

15.3.3 Gamma Knife

Radiosurgery is delivered using the Leksell^{R} Gamma Knife (LGK), which was invented by Dr. Lars Leksell in the early 1950s [5]. This device comprises 201-cobalt sources localized in a hemisphere around the patient's head so that all beams of gamma radiation converge onto the tumor.

15.3.4 Results

15.3.4.1 Tumor Control and Complications

Between 1992 and 2010, 189 patients were treated in Graz with prescription doses between 25 Gy and 80 Gy (Fig. 15.4), 177 of whom had complete data for analysis [2, 3, 6].The mean basal tumor diameter was 11.9 mm (range, 3.7–

19.2 mm), and the mean tumor height was 6.4 mm (range, 2.1–14.5 mm). Tumor regression was achieved in 94.4 % of patients. There were ten recurrences, after a median time of 13.6 months. The most important complication leading to secondary enucleation was neovascular glaucoma, developing in about 17 % of patients. Reduction of the prescription dose to ≤40 Gy and exclusion of ciliary body tumors decreased the rate of glaucoma without affecting the rate of tumor control. Others have reported similar ocular results [7–14].

15.3.4.2 Metastasis

Multivariate analysis indicates that 5 years after treatment, survival after stereotactic radiosurgery is not significantly worse than after enucleation [7, 15].

15.4 Fractionated Stereotactic Radiotherapy

Fractionated stereotactic radiotherapy combines the precision of stereotactic positioning with the radiobiological advantage of fractionation, thereby reducing long-term toxicity.

15.4.1 Treatment Planning

The size, shape, and location of the tumor are defined by ultrasonography, high-resolution CT, and MRI. A computerized 3-D model of the eye and tumor is generated. The CT and MRI images are fused so that they can be viewed in axial, sagittal, or coronal planes. During CT and MRI delineation, the patient wears the same head mask and ocular fixation system as for subsequent treatment. A total dose of 50–60 Gy is usually delivered in five fractions over a period of up to 10 days [16, 17]. The tumor is treated with a safety margin of 2.0–2.5 mm in all directions. Additionally, critical structures (lens, optic nerve, anterior eye segment, lacrimal gland) are contoured for treatment plan optimization, which includes dose-volume histograms.

15.4.2 Procedure

Immobilization of the patient's head is usually achieved by means of a noninvasive, rigid, thermoplastic mask and bite block, which are individually prepared for each patient (Fig. 15.5). This mask system is attached to the patient's couch. The patient is asked to look at a blinking fixation light with the diseased eye or the fellow

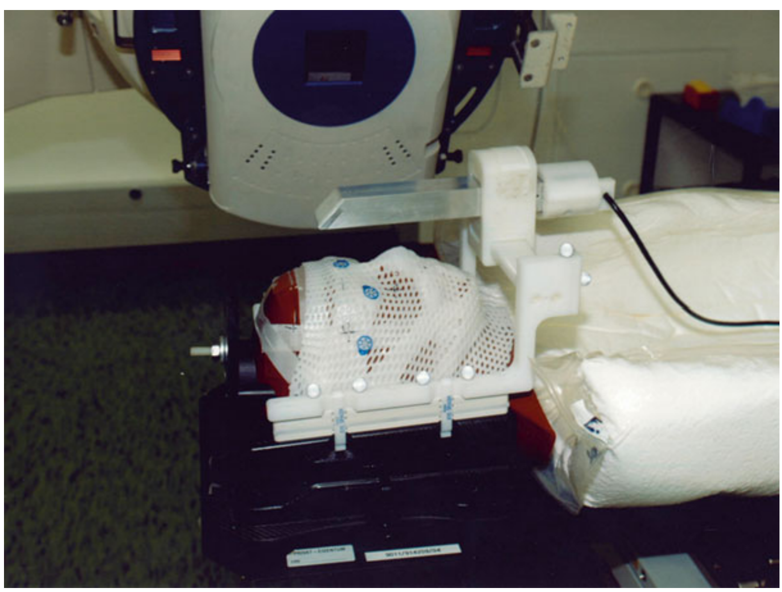

Fig. 15.5 Set-up prior to LINAC SRT treatment depicting thermoplastic head mask and ocular fixation system within alloy tube consisting of a mirror, a fixation light, infrared illumination and TV camera

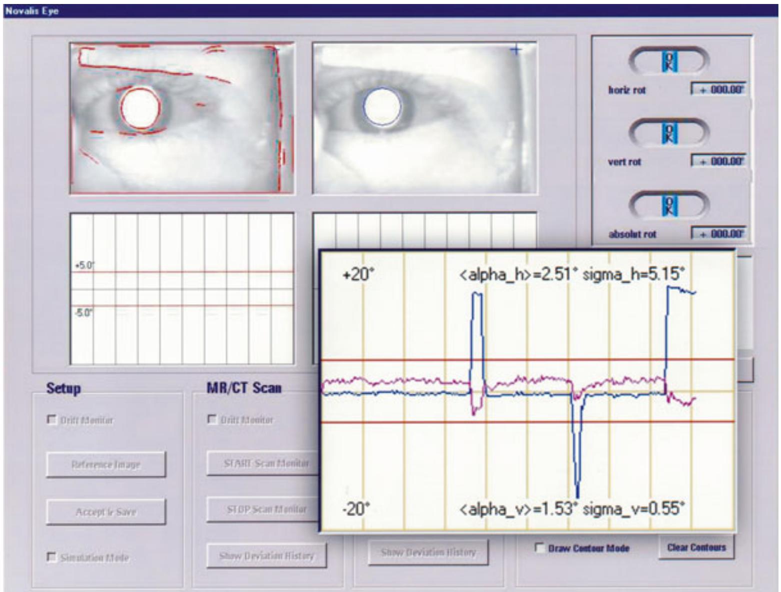

Fig. 15.6 Screen shot of software used for recording eye movements during delineation and LINAC irradiation

eye [16, 17]. The eye position is monitored using an infrared illumination system and mini-video camera. Any ocular deviations are detected by means of customized graphic software, and if these exceed calculated limits, the beam delivery is automatically interrupted until proper eye position is regained (Fig. 15.6) [18, 19]. At the start of each treatment session, a quality assurance test of the complete setup is undertaken, using a laser positioning system to ensure that the tumor is accurately placed at the focal point of treatment.

15.4.3 Linear Accelerator

Using a linear accelerator (LINAC), a single beam of photons is sequentially directed at the tumor from many different directions through multiple, non-coplanar arcs or collimated static fields. The beam entry is spread over the superior part of the head. This ensures that the dose to surrounding tissue is minimized. In most cases, the beam is collimated with a micro-multileaf collimator consisting of multiple pairs of tungsten leaves (mMLC) [20]. Computer software automatically adjusts the collimation in the mMLC for every angle of treatment according to the contour of the tumor viewed from at that angle (Fig. 15.7). A comparative treatment planning study showed levels of dose conformation to be similar to proton beam radiotherapy [21].

15.4.4 Results

15.4.4.1 Tumor Control and Complications

Between 1997 and 2007, 212 patients were treated with LINAC SRT in Vienna (Fig. 15.8). This therapy was selected for tumors “with” a thickness exceeding 7.0 mm or extending to within 3 mm of the optic disc or macula. Interim and latest results have been published [9, 16, 22, 23].

The tumors had a mean thickness of 4.8 mm (range, 2.5–11.5). The median follow-up time was 65 months. Local tumor control at 5 and 10 years was 96 and 93 %, respectively. Long-term side effects included retinopathy ($n=114$; 54 %), optic neuropathy ($n=107$; 50.5 %), cataract ($n=87$; 41 %), and secondary neovascular glaucoma ($n=46$; 22 %). Secondary enucleation was performed in 39 patients (18 %). Other groups using this technique have reported similar results [24–27]. Radiobiological rationale and experimental studies suggest that there is scope for clinical trials with a reduced dose per fraction and a higher number of fractions [28].

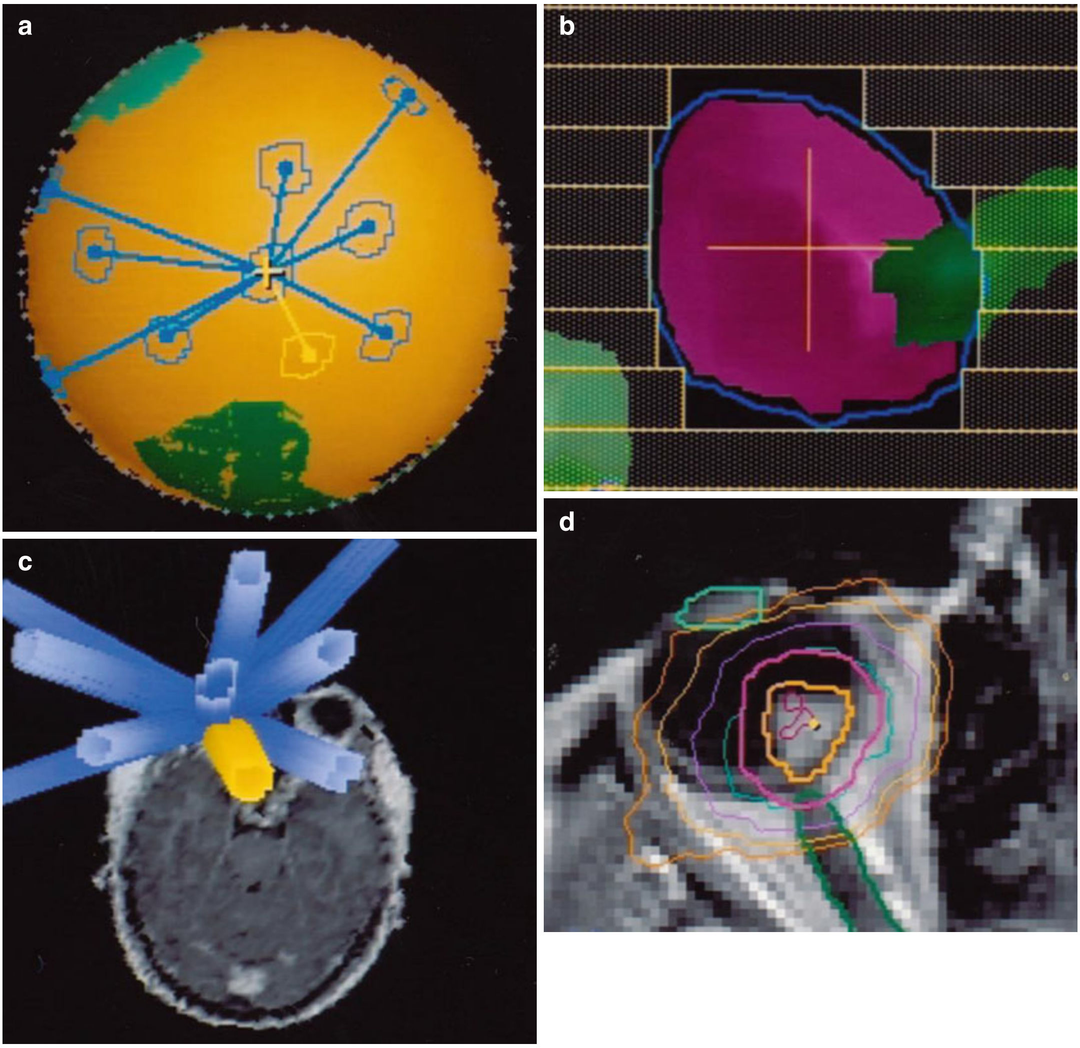

Fig. 15.7 Dosimetry plan for an juxtapapillary uveal melanoma (a & b) treated with 10 individually shaped static LINAC photon fields (c & d) (beam eye view, micro multileaf collimator)

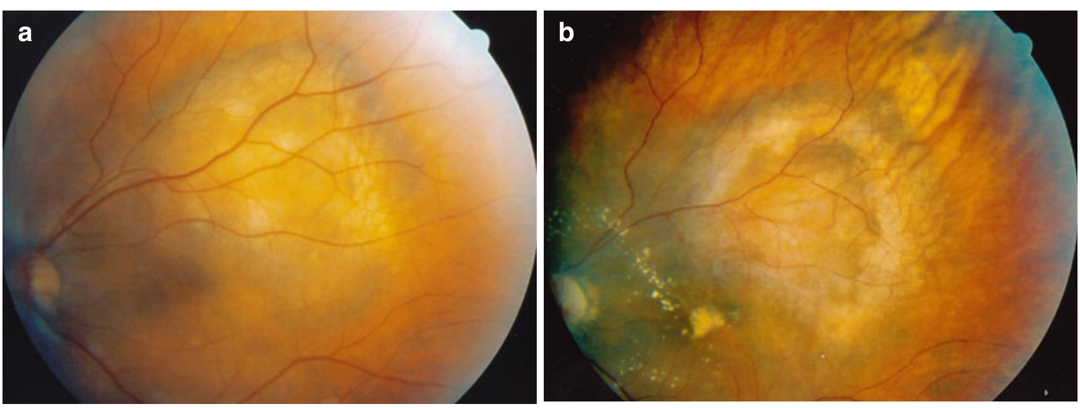

Fig. 15.8 Patient before (a) and after (b) LINAC SRT treatment

15.4.4.2 Metastasis

Thirty-two patients (15 %) developed metastases, and 22 of them died during the follow-up period.

15.5 Indications and Contraindications

Both forms of stereotactic radiotherapy are useful for tumors that are considered unsuitable for brachytherapy, because of either posterior location or large tumor size. With large tumors, stereotactic radiotherapy is contraindicated if the patient does not accept the increased chances of retinal detachment and neovascular glaucoma. Similarly, patients with posterior tumor extension must accept the increased likelihood of optic neuropathy and maculopathy. When there is a high risk of external eye and eyelid complications, careful counseling is required to ensure that the patient understands the implications regarding cosmesis and comfort.

For large uveal melanomas, stereotactic radiotherapy followed routinely by endoresection has been advocated as a means of improving local tumor control while avoiding neovascular glaucoma [29]. The scope of stereotactic radiotherapy in the treatment of iris melanoma has not yet been evaluated.

Conclusions

Radiosurgery and stereotactic radiotherapy are relatively new therapeutic modalities for uveal melanoma. As the techniques are still evolving, some still regard these modalities as investigational. Preliminary results suggest high rates of local tumor control, with conservation of vision depending on the size and location of the tumor. Therefore, stereotactic radiotherapy administered alone or in combination with other treatments can be considered for treatment of selected cases of uveal melanoma.

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